Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Starting position

Meaning in my words How would you explain starting position to a friend?	**How can you tell?** At 0 s the cart is beside the +5 m mark. Where does its position graph begin?
Example or sketch The cart starts at +5 m and moves 2 m left. What two positions would you mark?	**Non-example and why** “The clock starts at zero, so the cart starts at zero.” What do the two zeros mean?

Displacement

Meaning in my words How would you explain displacement to a friend?	**How can you tell?** Which two marks tell you the displacement even if the cart turns around?
Example or sketch Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement?	**Non-example and why** Someone adds both legs and gets 6 m. What did they find?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Final position

Meaning in my words How would you explain final position to a friend?	**How can you tell?** What must you combine with the starting position to locate the finish?
Example or sketch A cart starts at +5 m and moves 7 m left. Where would you draw it?	**Non-example and why** A partner reports 7 m because that is how far the cart traveled. What is missing?

Prediction range

Meaning in my words How would you explain prediction range to a friend?	**How can you tell?** A forecast reads 10 ± 2 km. Which positions are at its ends?
Example or sketch Where would you shade the allowed positions on a number line?	**Non-example and why** Someone circles only 10 km. What information disappears?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Mark the central forecast and shade its interval. Label one reason the interval could grow.

Choose zero, direction, and scale.



My marks mean

How wide should we search?

05

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Read forecast bounds without inventing a probability.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

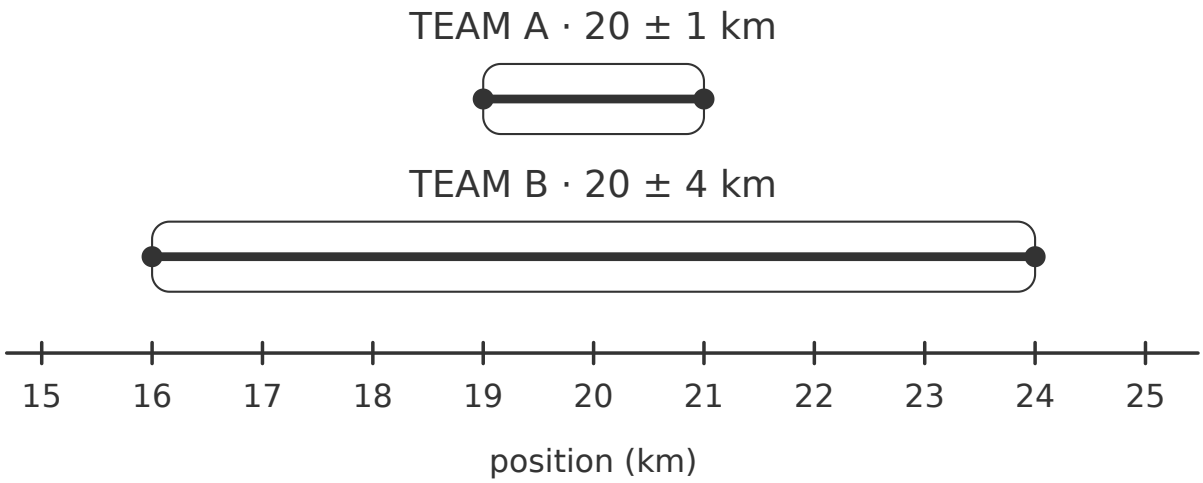
Label both search bands with their endpoints. Recommend which report needs a wider telescope search.

WEATHER-DESK PRESENTER ☐

Present both forecast bands. Explain which search is wider and what the bands do not tell us about impact.

Point to a clue. Keep your first marks; add the update using the color key.

Two XJ teams predict a center of **20 km**. Team A gives **±1 km**; team B gives **±4 km**. A telescope has one narrow field.



01 OUR FIRST CALL

What search area should the crew prepare for? Use both sets of endpoints.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the editor's new label. What can you report from these bands?

Switch speakers. Does a wider band tell us the chance of impact?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks